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Method for controlling flash memory module and associated flash memory controller and memory device

Abstract

The present invention provides a method for controlling a flash memory module. The flash memory module includes a plurality of dies, each die includes a plurality of blocks, each block includes a plurality of pages, and the method includes the steps of: selecting a super block, wherein the super block includes a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a flash memory.

2. Description of the Prior Art

(2) In the flash memory module, a block is used as an erase unit, that is, the erase operation can only be performed in units of blocks, and data of memory units of all pages within the block will be erased together. However, for some flash memory modules, if the erase operation is performed on a block where some pages have not yet been written, it may affect the subsequent write operation of this block, thereby reducing the quality of the data stored in the block.

SUMMARY OF THE INVENTION

(3) It is therefore an objective of the present invention to provide a method for accessing the flash memory module, which can efficiently determine whether a block has unwritten page(s) before the block is erased, and redundant data is written into the page(s) where data has not been written, so as to solve the problems described in the prior art.

(4) According to one embodiment of the present invention, a method for controlling a flash

memory module is disclosed. The flash memory module comprises a plurality of dies, each die comprises a plurality of blocks, each block comprises a plurality of pages, and the method comprises: selecting a super block, wherein the super block comprises a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks.

(5) According to one embodiment of the present invention, a flash memory controller configured to access a flash memory module is disclosed. The flash memory module comprises a plurality of dies, each die comprises a plurality of blocks, each block comprises a plurality of pages, and the flash memory controller comprises: a read-only memory configured to store a program code, and a microprocessor configured to execute the program code to access the flash memory module. The microprocessor is configured to perform the steps of: selecting a super block, wherein the super block comprises a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks.

(6) According to one embodiment of the present invention, a memory device comprising a flash memory module and a flash memory controller is disclosed. The flash memory module comprises a plurality of dies, each die comprises a plurality of blocks, each block comprises a plurality of pages. The flash memory controller, configured to access the flash memory module, and the flash memory controller is configured to perform the steps of: selecting a super block, wherein the super block comprises a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks.

(7) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating a memory device according to an embodiment of the present invention.

(2) FIG. 2 is a diagram of a super block according to an embodiment of the present invention.

(3) FIG. 3 shows a block including a plurality of pages.

(4) FIG. 4 is a flowchart of a control method of a memory device according to one embodiment of the present invention.

DETAILED DESCRIPTION

(5) FIG. 1 is a diagram illustrating a memory device **100** according to an embodiment of the present invention. The memory device **100** includes a flash memory module **120** and a flash memory controller **110**, wherein the flash memory controller **110** is arranged to access the flash memory module **120**. The flash memory controller **110** includes a microprocessor **112**, a read only memory (ROM) **112M**, a control logic **114**, a buffer memory **116** and an interface logic **118**. The ROM **112M** is arranged to store a program code **112C**, and the microprocessor **112** is arranged to

execute the program code **112C** to control access of the flash memory module **120**. The control logic **114** includes an encoder **132**, a decoder **134**, a randomizer **136** and a de-randomizer **138**. The encoder **132** is arranged to encode data that is written into the flash memory module **120** to generate a corresponding parity (also known as an error correction code (ECC)), and the decoder **134** is arranged to decode data that is read from the flash memory module **120**. The randomizer **136** is used to randomize the data written to the flash memory module **120**, and the de-randomizer **138** is used to de-randomize the data read from the flash memory module **120**.

(6) In a general situation, the flash memory module **120** includes a plurality of flash memory chips, and each flash memory chip includes a plurality of blocks. A controller (e.g. the flash memory controller **110** that executes the program code **112C** through the microprocessor **112**) may copy, erase, and merge data for the flash memory module **120** with a block as a unit. In addition, a block can record a specific number of pages, wherein the controller (e.g. the flash memory controller **110** that executes the program code **112C** through the microprocessor **112**) may perform a data write operation upon the flash memory module **120** with a page as a unit. In other words, a block is the smallest erase unit in the flash memory module **120**, and a page is the smallest write unit in the flash memory module **120**.

(7) In practice, the flash memory controller **110** that executes the program code **112C** through the microprocessor **112** may utilize its own internal components to perform many control operations. For example, the flash memory controller **122** utilizes the control logic **114** to control access of the flash memory module **120** (more particularly, access at least one block or at least one page), utilizes the buffer memory **116** and/or a DRAM **140** to perform a required buffering operation, and utilizes the interface logic **118** to communicate with a host device **130**.

(8) In one embodiment, the memory device **100** may be a portable memory device such as a memory card which conforms to one of the SD/MMC, CF, MS and XD specifications, and the host device **130** is an electronic device able to be connected to the memory device **100**, such as a cellphone, a laptop, a desktop computer, etc. In another embodiment, the memory device **100** can be a solid state drive (SSD) or an embedded storage device conforming to the universal flash storage (UFS) or embedded multi-media card (EMMC) specifications, and can be arranged in an electronic device. For example, the memory device **100** can be arranged in a cellphone, a watch, a portable medical testing device (e.g. a medical wristband), a laptop, or a desktop computer. In this case, the host device **130** can be a processor of the electronic device.

(9) FIG. 2 is a diagram of a super block according to an embodiment of the present invention. As shown in FIG. 2, it is assumed that the flash memory module **120** includes four dies (die 1 to die 4), and each die includes a plurality of blocks B1-BK. At this time, the microprocessor **112** can configure blocks belonging to different data planes or different dies within the flash memory module **120** as a super block to facilitate management of data access. In this embodiment, there is only one data in one die, but the invention is not limited to this. As shown in FIG. 2, die 1 to die 4 respectively include blocks B1-BK, and the microprocessor **112** can configure the block B1 of each die as a super block **202**, and configure the block B2 of each die as a super block **204**, and so on. The operation of the flash memory controller **110** in accessing the super blocks **202** and **204** is similar to accessing the general blocks. For example, the super block **202** itself is an erase unit. That is, although the four blocks B1 included in the super block **202** can be erased separately, the flash memory controller **110** will definitely erase the four blocks B1 together.

(10) FIG. 3 is a diagram illustrating a block **300**, wherein the block **300** can be any one of the blocks B1-BK shown in FIG. 2, and the block **300** includes a plurality of pages, such as 448 pages in the figure. When writing data to the super block **202**, the first page P1 of the block B1 of the die 1, the first page P1 of the block B1 of the die 2, the first page P1 of the die 3 and the first page P1 of the block B1 of the die 4 can be sequentially written. Then, the data is written into the second page P2 of the block B1 of die 1, the second page P2 of the block B1 of the die 2, . . . , and so on. In other words, the flash memory controller **110** writes data to the first page P1 of each block B1 in

the super block **202**, and then writes data to the second page P2 of each block B1 in the super block **202**. The super block is a collection block logically set by the flash memory controller **110** for the convenience of managing the flash memory module **120**, not a physical collection block. In addition, when performing garbage collection, calculating the effective pages of a block, and calculating the length of writing a block, calculations can also be performed in units of super blocks.

(11) In this embodiment, the flash memory module **120** is a three-dimensional NAND-type flash memory module, in which each block is composed of a plurality of word lines, a plurality of bit lines and a plurality of memory cells. Since the three-dimensional NAND flash memory architecture is well known to a person skilled in the art, no further explanation is given in the specification.

(12) FIG. **4** is a flowchart of a control method of the memory device **100** according to one embodiment of the present invention. In Step **400**, the flow starts, and the memory device is powered on. In Step **402**, the microprocessor **112** in the flash memory controller **110** determines whether there is currently a super block that needs to be erased, if yes, the flow enters Step **404**; and if not, the flow stays at Step **402** and continues to determine whether there is currently one super block that need to be erased. For example, the microprocessor **112** can record in the buffer memory **116** which super blocks in the flash memory module **120** are currently in an erasable state (that is, all the data stored in the super blocks is invalid data), and when the microprocessor **112** needs to write data from the host device **130** to the flash memory module **120**, or the microprocessor **112** needs to perform a garbage collection operation on some blocks in the flash memory module **120**, or microprocessor **112** requires a blank block for data writing, the microprocessor **112** determines that there is currently a super block that needs to be erased to generate at least one blank super block for subsequent use, and the microprocessor **112** selects at least one of the super blocks in the erasable state. In the following description, it is assumed that the super block **202** needs to be erased.

(13) In step **404**, the microprocessor **112** selects a block from the super block **202**, such as the block B1 of die 1. In step **406**, the microprocessor **112** determines whether the selected block is a full block or a blank block (empty block), if yes, the flow enters Step **412**; and if not, the flow enters Step **408**. Specifically, the microprocessor **112** can read the first page P1 of the selected block to determine whether the first page P1 has data stored therein. For example, because the microprocessor **112** will obtain logical value “1” when reading the blank page, the microprocessor **112** can determine whether the proportion of logical values “1” in the content of the first page P1 is higher than a threshold value, to determine whether the first page P1 has data. In another example, the microprocessor **112** can read the content of the spare region of the first page P1 to determine whether data has been written into the first page P1, that is, if the spare region of the first page P1 includes valid data, it is determined that the page P1 has data. If the spare area of the first page P1 only includes invalid data (for example, all the contents correspond to logical value “1”), it is determined that the first page P1 is a blank page. Then, the microprocessor **112** reads the last page P448 of the selected block to determine whether data has been written into the last page P448. If both the first page P1 and the last page P448 have data, it is determined that the selected block is a full block and the flow enters Step **412**. If both the first page P1 and the last page P448 are blank, it is determined that the selected block is a blank block and the flow also enters Step **412**. In addition, if data is written into the first page P1, but the last page P448 is blank, it is determined that the selected block is not a full block or a blank block (that is, a half-full block), and the flow enters Step **408**.

(14) In Step **408**, the microprocessor **112** uses a binary search method or any other suitable search method to search for the last page in the selected block that has stored data therein. In Step **410**, the microprocessor **112** writes dummy data to all blank pages located after the last page that has been written in the selected block, so that all pages of the selected block have written data to become a

full block, wherein the dummy data can be invalid data with any data type and not corresponding to any logical address of the host device **130**. Taking FIG. **3** as an example, if the microprocessor **112** determines that the last page with data written in the block **300** is PA, the microprocessor **112** will write dummy data to all of the pages P(A+1) PK.

(15) In Step **412**, the microprocessor **112** determines whether all blocks in the super block **202** have been processed (that is, all blocks have passed Step **406**). If yes, the flow enters Step **416**; and if not, the flow enters Step **414**.

(16) In Step **414**, the microprocessor **112** selects the next unprocessed block in the super block **202**, such as block B1 of die 2, and then the flow goes back to Step **406**.

(17) In Step **416**, since all blocks in the super block **202** are currently full blocks or blank blocks, the microprocessor **112** can generate a control signal to the flash memory module **120** to erase all blocks in the super block **202**, so that all blocks in the super block **202** become blank blocks for subsequent use.

(18) In the embodiment of FIG. **4**, since all blocks in the super block **202** are already full blocks or blank blocks when erasing, it can avoid the problem in the prior art that the subsequent writing operation of this block is affected by erasing a half-full block.

(19) In addition, because some flash memory modules do not allow double programming, that is, pages that have already been written are not allowed to be written again before erasing, so the above Step **408** and Step **410** can avoid the problem that pages may be permanently damaged due to double programming.

(20) In another embodiment, if the flash memory module **120** allows double programming, Step **408** can be removed from the flow of FIG. **4**, and Step **410** is modified to the microprocessor **112** directly writing dummy data into all pages in the selected block, that is, the pages P1-P448 in FIG. **3** will be written with dummy data in sequence.

(21) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A method for controlling a flash memory module, wherein the flash memory module comprises a plurality of dies, each die comprises a plurality of blocks, each block comprises a plurality of pages, and the method comprises: selecting a super block, wherein the super block comprises a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks; wherein the step of for each first block in the super block, determining whether the first block is the full block or the blank block comprises: reading a first page of the first block to determine whether the first page has written data; reading a last page of the first block to determine whether the last page has written data; if both the first page and the last page have written data, determining the first block to be the full block; and if both the first page and the last page do not have written data, determining the first block to be the blank block.

2. The method of claim 1, wherein the step of selecting the super block comprises: when the flash memory module requires one or more blank blocks for data writing operations, the super block is selected from a plurality of super blocks with erasable state.

3. The method of claim 1, wherein the step of if the first block is not the full block or the blank block, writing the dummy data to the first block so that the first block becomes the full block

comprises: starting from a first blank page of the first block, writing the dummy data into all blank pages of the first block, so that the first block becomes the full block.

4. The method of claim 1, wherein the step of if the first block is not the full block or the blank block, writing the dummy data to the first block so that the first block becomes the full block comprises: starting from a first page of the first block, writing the dummy data into all of the pages of the first block, so that the first block becomes the full block.

5. A flash memory controller, wherein the flash memory controller is configured to access a flash memory module, the flash memory module comprises a plurality of dies, each die comprises a plurality of blocks, each block comprises a plurality of pages, and the flash memory controller comprises: a read-only memory, configured to store a program code; a microprocessor, configured to execute the program code to access the flash memory module; wherein the microprocessor is configured to perform the steps of: selecting a super block, wherein the super block comprises a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks; wherein the step of for each first block in the super block, determining whether the first block is the full block or the blank block comprises: reading a first page of the first block to determine whether the first page has written data; reading a last page of the first block to determine whether the last page has written data; if both the first page and the last page have written data, determining the first block to be the full block; and if both the first page and the last page do not have written data, determining the first block to be the blank block.

6. The flash memory controller of claim 5, wherein the step of selecting the super block comprises: when the flash memory module requires one or more blank blocks for data writing operations, the super block is selected from a plurality of super blocks with erasable state.

7. The flash memory controller of claim 5, wherein the step of if the first block is not the full block or the blank block, writing the dummy data to the first block so that the first block becomes the full block comprises: starting from a first blank page of the first block, writing the dummy data into all blank pages of the first block, so that the first block becomes the full block.

8. The flash memory controller of claim 5, wherein the step of if the first block is not the full block or the blank block, writing the dummy data to the first block so that the first block becomes the full block comprises: starting from a first page of the first block, writing the dummy data into all of the pages of the first block, so that the first block becomes the full block.

9. A memory device, comprising: a flash memory module, wherein the flash memory module comprises a plurality of dies, each die comprises a plurality of blocks, each block comprises a plurality of pages; and a flash memory controller, configured to access the flash memory module; wherein the flash memory controller is configured to perform the steps of: selecting a super block, wherein the super block comprises a plurality of first blocks respectively located in the plurality of dies; for each first block in the super block, determining whether the first block is a full block or a blank block; and if the first block is not the full block or the blank block, writing dummy data to the first block so that the first block becomes the full block; and erasing the plurality of first blocks in the super block, so that the plurality of first blocks become a plurality of blank blocks; wherein the step of for each first block in the super block, determining whether the first block is the full block or the blank block comprises: reading a first page of the first block to determine whether the first page has written data; reading a last page of the first block to determine whether the last page has written data; if both the first page and the last page have written data, determining the first block to be the full block; and if both the first page and the last page do not have written data, determining the first block to be the blank block.

10. The memory device of claim 9, wherein the step of selecting the super block comprises: when

the flash memory module requires one or more blank blocks for data writing operations, the super block is selected from a plurality of super blocks with erasable state.

11. The memory device of claim 9, wherein the step of if the first block is not the full block or the blank block, writing the dummy data to the first block so that the first block becomes the full block comprises: starting from a first blank page of the first block, writing the dummy data into all blank pages of the first block, so that the first block becomes the full block.

12. The memory device of claim 9, wherein the step of if the first block is not the full block or the blank block, writing the dummy data to the first block so that the first block becomes the full block comprises: starting from a first page of the first block, writing the dummy data into all of the pages of the first block, so that the first block becomes the full block.
